
Clean Accuracy Is the Wrong Lens: Reading Invariance and Capacity from Digit Networks

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Abstract

On MNIST, modern networks are separated by a fraction of a percentage point of test accuracy, which invites the conclusion that the architecture no longer matters. We argue the opposite: clean accuracy *hides* what each network computes and needs, and two cheap readouts—perturbation robustness and per-layer linear separability—recover it. (i) A panel spanning an MLP, a small CNN, and the MNIST record-holder model differs by only 1.4 percentage points on clean data but by ~ 46 pp under a 4-pixel shift: the architectural gap is an *invariance* gap, invisible on clean accuracy and ordered exactly by spatial structure. (ii) A depth ablation shows clean accuracy and per-layer linear separability both saturate at ~ 4 convolutional blocks—beyond which the representation is already linearly separable, so no non-linearly-separable (“XOR-like”) classification work remains—yet translation robustness keeps improving with depth, giving a concrete capacity rule: few layers suffice for clean accuracy, more buy invariance. (iii) The networks also carry recoverable latent structure: an unsupervised style variant (crossed vs. uncrossed “7”) is linearly decodable *and* generatively steerable, and depth makes internal class prototypes clean. The unifying message is that robustness and linear separability, not clean accuracy, are the informative axes for reading digit networks.

1 Introduction

MNIST [5] is often treated as solved: the reproducible record is 99.87% [2] and competing architectures cluster within a percentage point. This near-saturation makes clean test accuracy almost useless for *comparing* or *understanding* digit networks. We show that two inexpensive measurements recover the information that clean accuracy hides, and that they tell a consistent story.

Our contributions are three readouts and one unifying claim. **(1) Model differences are invariance differences**, readable from perturbation robustness rather than clean accuracy. **(2) Depth past a linear-separability “knee” buys robustness, not accuracy**—a concrete, actionable capacity rule obtained from a depth ablation plus per-layer linear probes [1]. **(3) Digit networks carry recoverable latent structure**: an unsupervised style variant is both linearly decodable and generatively steerable, and depth yields clean internal class prototypes (via activation maximization [3, 8, 7]). The unifying claim: *robustness and linear separability are the informative axes; clean accuracy is the wrong lens*.

2 Setup

All models are trained on MNIST with one backbone family per experiment. The comparison panel is an **MLP** (784–256–128–10), the 2-conv **SimpleCNN**, and **M3**, the single deepest sub-model (10 conv layers, no pooling) of the An et al. record holder [2]. The depth study uses a **DepthCNN** family of 1–8 identical conv blocks. We perturb the test set by integer translation and rotation (torchvision affine), fit linear probes (logistic regression) on intermediate features, and reconstruct “canonical”

Table 1: Panel (3 seeds): clean accuracy barely separates the models; robustness does.

Model	Clean acc	Acc @ 4px shift	Acc @ 20° rot	variant-7 probe (base 57)
MLP	98.08	39.7	87.1	80.1
SimpleCNN	99.12	71.7	93.4	79.2
M3 (record)	99.47	85.8	95.3	79.9

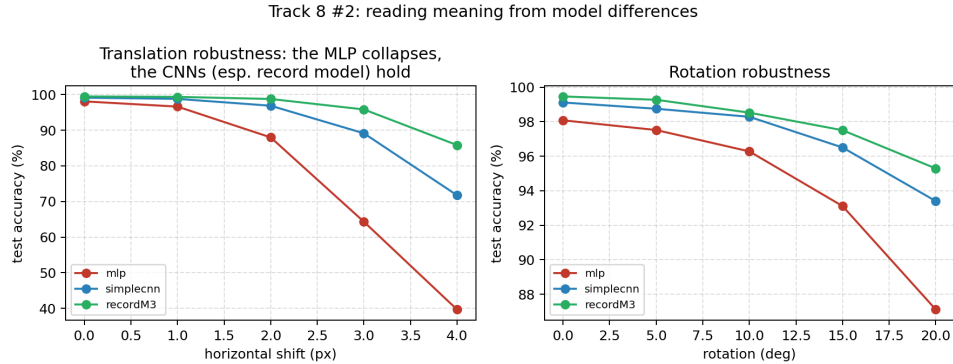


Figure 1: Reading 1. Clean (0 shift) the three models coincide at $\sim 98\text{--}99\%$; under perturbation they fan out by tens of points, ordered by spatial structure. The architectural difference is an invariance gap.

class images by gradient ascent on the input with total-variation/blur regularization. All numbers are means over 3 seeds.

3 Reading 1: model differences encode invariance

Table 1 and Figure 1 report the panel. On clean MNIST the three architectures span just 1.4 pp (98.08/99.12/99.47)—they look nearly equivalent. Under a 4-pixel shift they span ~ 46 pp (39.7/71.7/85.8): the MLP, with no spatial prior, collapses; the convolutional prior recovers much of it; and depth without pooling (M3) recovers most. The ordering $\text{MLP} \ll \text{SimpleCNN} \ll \text{M3}$ is identical for translation and rotation. Thus “model A beats model B ” is best read as “ A computes more invariance,” a statement clean accuracy cannot make. **Honest null:** an unsupervised style-variant probe (§5) reaches $\sim 80\%$ on *all three* representations (base 57%)—structure preservation is architecture-independent, so the discriminating axis is robustness, not representation richness.

4 Reading 2: depth past the linear-separability knee buys robustness, not accuracy

A classic fact motivates the question: a single linear unit cannot compute XOR, the smallest non-linearly-separable function [6]; depth exists to implement such functions. So if a network’s representation is already *linearly* separable for the task at layer L , layers beyond L do no further non-linearly-separable (“XOR-like”) classification work, and one might cut them. We test this directly (Figure 2). Training DepthCNNs of 1–8 blocks, clean accuracy saturates by ~ 4 blocks (99.06 $\rightarrow$ 99.27 from block 3 to 8). A per-layer linear probe on the 8-block net confirms the cause: a *linear* classifier already reaches 99.1% by block 4 (final 99.5%), i.e. the representation is linearly separable there. *However*, translation robustness keeps climbing well past the knee: $\text{acc@4px} = 91.5$ (d3) $\rightarrow$ 95.5 (d4) $\rightarrow$ 98.5 (d8). The resulting **capacity rule**: for clean MNIST accuracy ~ 4 blocks suffice and deeper is wasted; if invariance/robustness matters, extra depth keeps paying. This is the same axis as Reading 1—added capacity buys robustness, not clean accuracy.

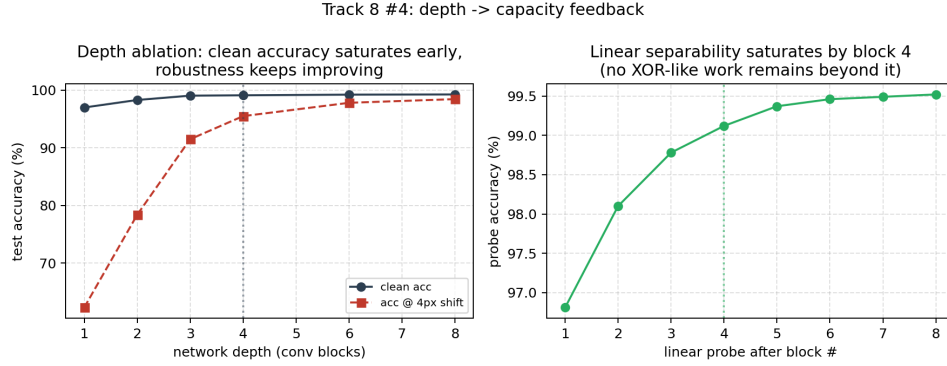


Figure 2: Reading 2. Left: clean accuracy saturates by ~ 4 blocks while 4-px-shift robustness keeps rising with depth. Right: a per-layer linear probe saturates by block 4—no “XOR-like” work remains beyond it, so the leftover gains from depth are invariance, not separability.

8 #1: canonical digits per model -- depth yields smooth, prototype-like templates (record M3/M5/M7) vs high-frequency texture (shallow Simple)



Figure 3: Reading 3. “Canonical” class images by activation maximization. The deep record sub-models (M3/M5/M7) form smooth, prototype-like digits; the shallow CNN forms high-frequency texture.

5 Reading 3: recoverable latent structure and clean prototypes

Decodable + generative style variant. A plain 10-class CNN never sees style labels, yet a crossed-vs-uncrossed “7” variant (defined from the skeleton graph [4]) is linearly decodable from its embedding (83.8% probe vs. 57% base). The axis is not merely decodable but *generative*: reconstructing a “7” while pushing its embedding along the probe direction moves the variant score monotonically ($-8.2 \rightarrow +25.0$) and adds/removes crossbar structure—so the network can be driven to render either variant.

Depth yields clean prototypes. Activation-maximizing each class (Figure 3) shows the deep record sub-models render smooth, prototype-like canonical digits (mean total variation ~ 0.08) where the shallow SimpleCNN renders high-frequency texture that barely resembles a digit (TV ~ 0.5). Smoothness tracks *depth*, not kernel size; the three record sub-models’ templates differ, consistent with their ensemble’s decorrelation.

6 Discussion

The three readings agree on one axis. Clean accuracy is saturated and near-constant across architectures and depths; the variation that *does* carry architectural meaning—which model generalizes under shift, whether extra depth helps, what the network has internalized—shows up in robustness and in linear separability, not in the headline number. Practically: compare models by perturbation

robustness, not clean accuracy; choose depth by the linear-separability knee unless robustness is the goal; and remember that a discriminative digit net carries decodable, steerable generative structure.

Limitations. This is MNIST and a small model family; the individual tools (activation maximization, linear probes, robustness evaluation) are established—our contribution is the *synthesis* and the concrete digit-network reading, not new machinery. The capacity rule is a demonstrated trend on this task, not a theorem, and “no XOR-like work remains” is an operational statement about linear-probe saturation, not a circuit-level claim.

7 Conclusion

On a near-saturated benchmark, clean accuracy is the wrong lens. Perturbation robustness reveals that model differences are invariance differences; per-layer linear separability reveals that depth past a knee buys robustness rather than accuracy; and activation maximization plus linear probes reveal recoverable, steerable latent structure. Read digit networks by what they are robust to and by where they become linearly separable—not by the last decimal of test accuracy.

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Reproducibility: `scripts/run_track8_model_panel.py` (Reading 1), `run_track8_depth.py` (Reading 2), `run_track8_introspect_best.py` + `run_introspection_reconstruct.py` (Reading 3). Raw results: `track8_panel_results.json`, `track8_depth_results.json`; full notes in `cnn_introspection_findings.md`.

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